

# Breathing In, Breathing Out — Paper A

MEDIUM • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- A calculator is allowed, but every number here is designed to work without one.
- Where a question asks about air moving in or out, your answer must follow the chain in order: **muscles change the volume, the volume change causes a pressure change, air then flows down the pressure gradient**. A muscle never sucks or pushes air by itself.
- The mark for each question is shown in square brackets on the right.
- You may need: **1000 cm<sup>3</sup> = 1 dm<sup>3</sup>**.

## SECTION 1 – AIR IN, AIR OUT

- 1.** Complete the table to describe quiet breathing. Each empty box is worth one mark. [6]

|                   | Breathing in           | Breathing out    |
|-------------------|------------------------|------------------|
| Diaphragm         | contracts and flattens |                  |
| Ribs              |                        | move down and in |
| Volume of thorax  |                        |                  |
| Pressure in lungs | falls                  |                  |
| Air               |                        | flows out        |

- 2.** Name the structure described in each case.

- (a)** The tube from the throat down towards the lungs, held open by rings of cartilage. [1]

- (b)** The two tubes that carry air from that tube, one into each lung. [1]

- (c)** The tiny air sacs where gases are exchanged with the blood. [1]

- (d)** The sheet of muscle below the lungs that flattens when you breathe in. [1]

- 3.** Breathing and respiration are not the same thing. Write one sentence about each to say what it is and where in the body it happens. [2]

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*One sentence each. If both your sentences mention the lungs, one of them is wrong.*

SECTION 2 – WHERE THE SWAP HAPPENS

- 4.** Give three features of an alveolus that make gas exchange fast. [3]

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- 5.** Answer these three questions about diffusion at an alveolus.

- (a)** Name the gas that diffuses from the air in the alveolus into the blood. [1]

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- (b)** Name the gas that diffuses in the opposite direction. [1]

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- (c)** State what both of these movements have in common. [1]

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- 6.** The inside of an alveolus is coated with a thin film of liquid. Explain why a gas exchange surface has to be moist. [2]

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SECTION 3 – THE REACTION, AND THE EVIDENCE FOR IT

- 7.** Answer these questions about aerobic respiration.

**(a)** Complete the word equation: [2]

glucose + \_\_\_\_\_ → \_\_\_\_\_ + water

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**(b)** Name the part of the cell in which the reaction takes place. [1]

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*Write that energy is released, never that it is made or produced.*

**8.** Name the liquid used to test for carbon dioxide, and state what you would see when a person breathes out through it. [2]

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**9.** Complete the table of gas percentages. Each empty box is worth one mark. [3]

| Gas            | Inspired air | Expired air |
|----------------|--------------|-------------|
| Oxygen         | 21%          |             |
| Carbon dioxide |              | 4%          |
| Nitrogen       | 78%          |             |

**10.** State the two things about a person's breathing that change when they start to exercise. [2]

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*End of paper. Check every answer has a unit or a form the question actually asked for.*